Draft v0.1 — charts use illustrative sample values until the notebooks run

Skills Map: closing the gap between your degree and the job ad

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36118 Applied NLP
Assessment 2A, group poster, Spring 2026

> *We put subject outlines, job ads and resumes into one skills vocabulary, so a student can see exactly which skills a role needs that they lack, and which subject fills each gap.*

1 The problem

Subject outlines speak academic. Job ads speak industry. Nothing maps one to the other, so students guess which electives to take and what to put on a resume.

> *"Apply and evaluate* **machine learning** *algorithms on real data and communicate the results."*
> UTS Handbook, subject learning objective (representative wording)

> *"Proven experience building and deploying* **machine learning** *models in production on AWS or Azure."*
> LinkedIn ad, Data Scientist, Sydney

Same skill, different words, and only the ad asks for cloud. The app makes that difference visible. It takes the teaching team's study planner idea from the timetable to the job market.

2 Users and goals

Elective chooser
Which subject moves me closest to the role I want?

Job seeker
What does my resume lack for this ad?

Career switcher
How far is my profile from a new role?

- Upload a resume, or tick the subjects you have taken
- Pick a target role and city
- See your skill gap ranked by demand
- Get electives ranked by how much gap they close
- Ask a chat that answers from real ads

Ethics

Resumes stay in memory, never on disk. Names, emails and phone numbers are masked before any LLM call. Ads are used under the dataset licence. Results are labelled as suggestions, not career advice.

References

European Commission (2024). ESCO v1.2 skills taxonomy.
Kon, A. (2024). LinkedIn Job Postings 2023–2024. Kaggle.
Reimers, N. and Gurevych, I. (2019). Sentence-BERT. EMNLP.
Lewis, P. et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP. NeurIPS.

3 Data

124k
LinkedIn job postings 2023–24, Kaggle
the demand side

13.9k
ESCO skills v1.2, European Commission
the shared vocabulary, with aliases

~180
UTS subjects, Handbook
descriptions and learning objectives

2.4k
Resumes, Kaggle Resume Dataset
test set for resume parsing

O*NET
Occupation profiles, release 29
role to skill priors

Seek
Australian ads, small sample
Sydney view, if site terms allow

4 Pipeline

Input
subject outline, job ad or resume; PDF resumes parsed with pypdf

Clean
deduplicate ads, strip boilerplate, split sentences

Extract skills
three ways, compared: ESCO dictionary and regex, spaCy NER, LLM

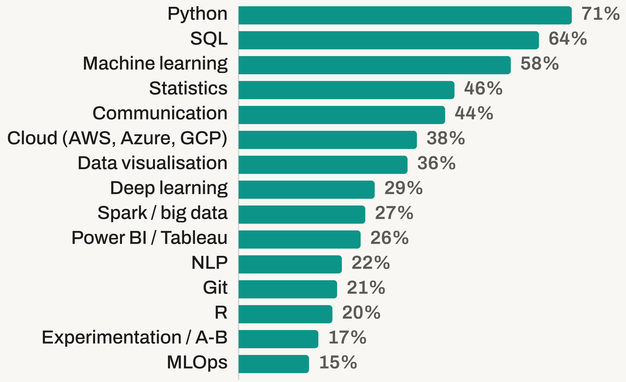
Normalise
map aliases to one ESCO id, so "ML" and "machine learning" agree

Match
two ways, compared: TF-IDF cosine versus sentence embeddings

Output
gap chart, ranked electives, demand trend, RAG chat over ads

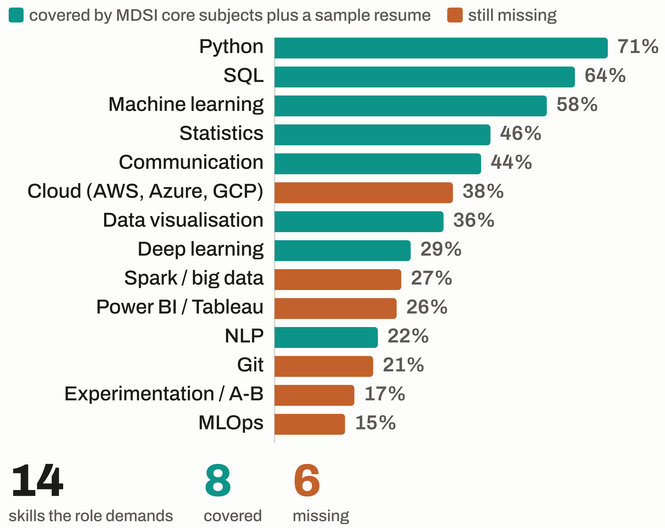
Copper steps are where a traditional and a newer method run side by side and are scored (block 8).

5 What the market asks for



Share of Data Scientist ads in Australia that mention each skill. Python and SQL lead; communication outranks deep learning. LinkedIn postings 2023–24, ESCO dictionary extraction.

6 Preliminary result: one profile against one role



Target role: Data Scientist, Sydney. Bar length is demand; colour is coverage. The biggest gaps are Cloud, Spark, Power BI: tools, not theory.

7 Which subject fills each gap

Missing skill	Best matching subject	Matched objective
Cloud	36120 Advanced Machine Learning Applications	*deploy models as services*
Spark, big data	36114 Advanced Data Algorithms	*algorithms at scale*
Power BI, Tableau	36104 Data Visualisation and Narratives	*interactive dashboards*
Experimentation	36103 Statistical Thinking for Data Science	*design and test hypotheses*
MLOps	36105 iLab: Capstone Project	*ship a working pipeline*

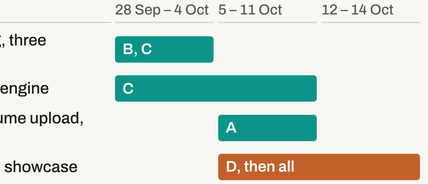
Ranked by cosine similarity between the missing ESCO skill and each subject's learning objectives.

8 Evaluation plan

Task	Traditional	Newer	Metric, test set
Skill extraction	ESCO dictionary + regex; spaCy NER in between	LLM: llama3.2 locally, GPT-4o-mini as check	F1 on 500 hand-labelled ads
Subject or resume to job matching	TF-IDF cosine	Sentence embeddings, all-MiniLM-L6-v2	Precision@5 on 50 team-labelled pairs
Job-ad chat	Keyword search + template answer	FAISS retrieval + LLM (RAG)	LLM-as-judge, 30 questions: faithfulness, relevance

A cheaper method stays in the app if it comes within 5 points of the newer one; the report shows both.

9 Plan to 14 October



Technical setup. GitHub repo AndyPham2004/ANLP-AT2-Skills-Map, Python 3.12 with uv, spaCy, sentence-transformers, FAISS, Streamlit; Ollama llama3.2 on a laptop, OpenAI API only for the accuracy check.